Case Study

Delhi Traffic Inefficiency & Spatial Congestion Mapping

An End-to-End Data Analytics Pipeline & Policy Framework

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Portfolio:
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Phase 1: ASK — The Urban Mobility Challenge

Problem: Fixed-time traffic signals in Delhi fail to adapt to dynamic vehicular surges, causing gridlocks and high economic loss.

Core Research Questions (BQ):

- **BQ1:** When do congestion peaks occur, and how severe are they?
- **BQ2:** What is the modal split among cars, buses, bikes, and heavy trucks?
- **BQ3:** How does traffic demand vary across Day Parts (Morning, Afternoon, Evening, Night)?
- **BQ4:** How heavily do daytime commercial trucks choke passenger traffic?

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Phase 2 & 3: PREPARE & PROCESS — Production Pipeline

Storage & Schema

Integrated SQLite DB (delhi_traffic.db) with production-grade logging.

SQL ETL Engine

Standardization using TRIM(), CAST(), COALESCE(), and dynamic CASE statements.

Automated Data Quality Gate

5-check audit (Nulls, Schema Types, Negative Values, Duplicates, Range Boundaries).

Feature Engineering

- Cyclical Time: Hour_Sin & Hour_Cos for ML readiness.
- Traffic Signals: Bus_To_Car_Ratio, Heavy_Vehicle_PCU_Share, Congestion_Index, and PCU_Lag_15Min.

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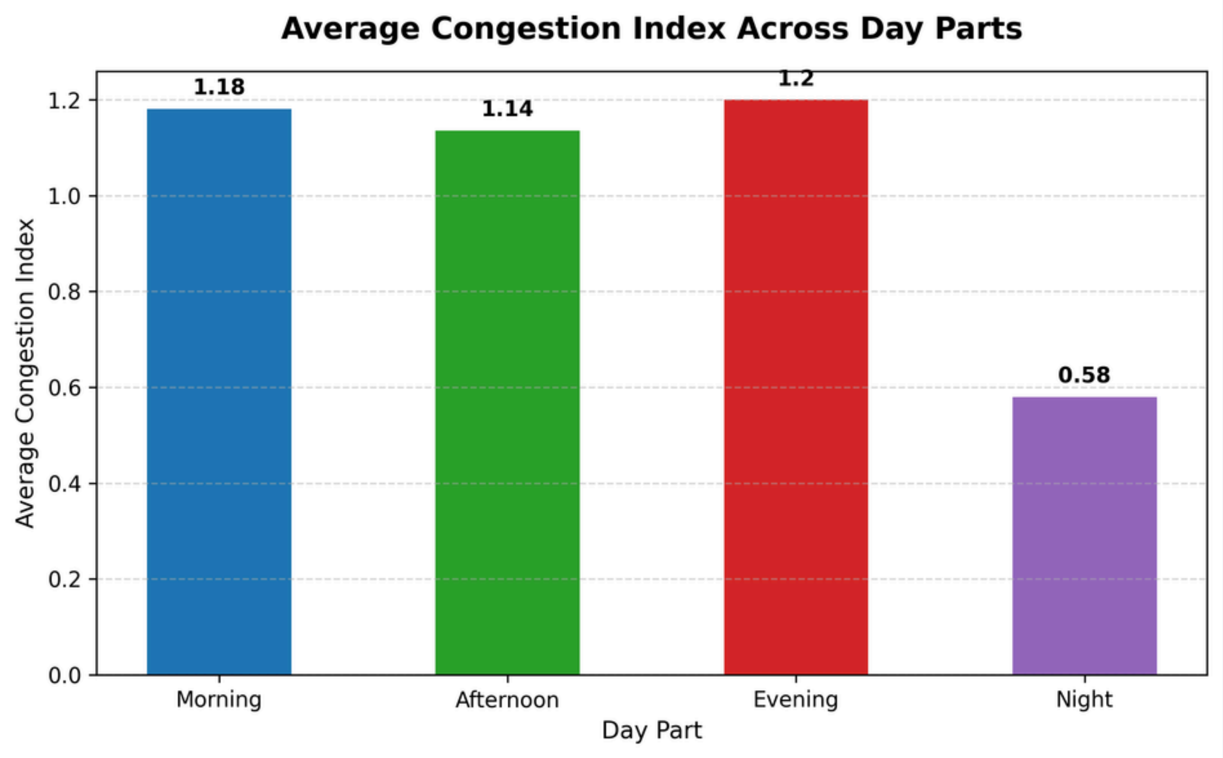
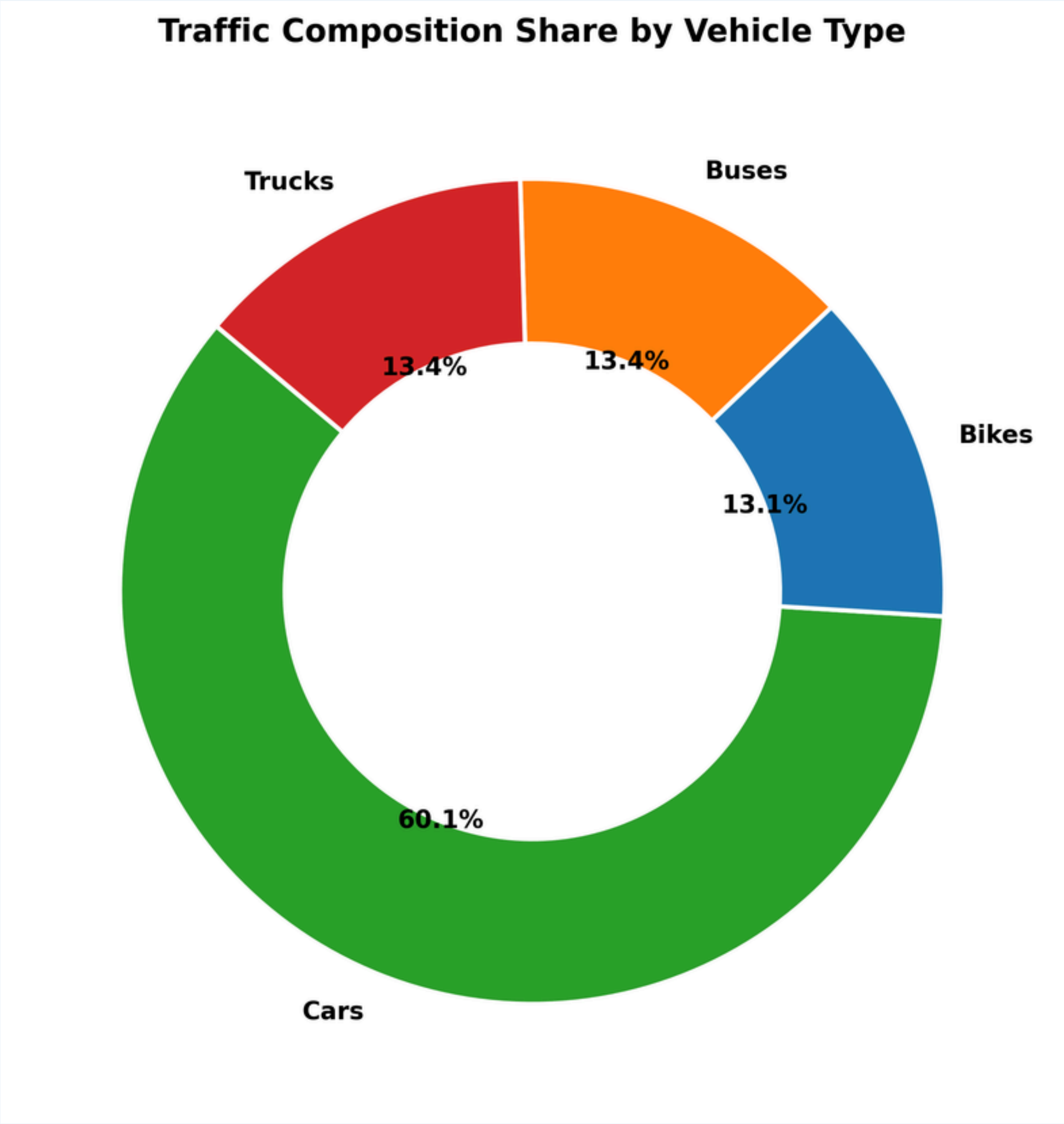
Phase 4: ANALYZE — Vehicle Fleet Composition & Congestion Spikes





- 60.1% — Private Cars Volume Share
- 1.20x — Evening Rush Congestion Index (Peak PCU ~246)
- 1.18x — Morning Rush Congestion Index (Peak PCU ~233)
- 0.58x — Night Window Congestion Index (Lowest Load Window)

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Phase 4: ANALYZE — Rigorous Statistical Validation

Hypothesis Test	Statistical Method	Key Result	Conclusion
Peak vs Off-Peak PCU	Two-Sample Welch's T-Test	$p<0.0001$	Reject H0: Significant PCU surge during rush windows.
Day Part Variance	One-Way ANOVA	$p<0.0001$	Reject H0: Variance across Day Parts is statistically real.
Bus Impact on PCU	Pearson Correlation (r)	$r=0.871$	Strong positive driver of total PCU score.
Truck vs Car Speed	Pearson Correlation (r)	$r=-0.628$	Heavy freight chokes passenger car throughput.

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Phase 5: SHARE — Dynamic Analytics Dashboard

- Built & Deployed an interactive Looker Studio (Google Data Studio) Executive Dashboard connected directly to the processed data mart (tableau_powerbi_dashboard_mart.csv).
- Interactive Controls: Dynamic dropdown filters for Day_Part (Morning, Afternoon, Evening, Night) and Rush_Hour_Category / Is_Peak_Hour for real-time slice-and-dice.
- Real-Time KPI Scorecards: Instant calculation of Total Monitored Vehicles (2.97M), Average PCU Score, Peak Bottleneck PCU (246), and Private Car Volume Share (60.1%).

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Phase 6: ACT — Actionable Policy & Engineering Roadmap

3-Tiered Implementation Plan:

1. **Short-Term (0–3 Months):**
    - Implement Dynamic Signal Control (extend green lights by 25–30% during Evening Rush 16:00–19:00).
    - Enforce Peak-Hour Freight Restrictions (Ban multi-axle trucks 07–10 AM & 04–09 PM).
2. **Medium-Term (3–12 Months):**
    - Demarcate Bus Rapid Transit (BRT) priority lanes to isolate r=0.871 queue bottlenecks.
    - Establish High-Occupancy Vehicle (HOV-2+) lanes to tackle 60.1% single-car volume.
3. **Long-Term (12–24 Months):**
    - Shift heavy commercial logistics exclusively to the Night Window (22:00–05:00, where index drops to 0.58).

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Conclusion & Future Scope

Key Takeaways:

- Successfully built a repeatable, production-grade Data Pipeline addressing real-world metropolitan traffic inefficiencies.
- Validated policy frameworks using advanced statistical inference and SQL data marts.

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Thank You —
Let's Connect

Open for Questions & Technical Discussion.

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